# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

**Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry**

| **Total Marks: 65** | **ANSWER KEY (Version 1)** | **Time: 2h 30m** |
|---|---|---|

## SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

| # | Ans | Correct Option | Explanation |
|---|---|---|---|
| **1** | **B** | **ab + ac − ad** | This is the **distributive property**: a(b + c − d) = ab + ac − ad. Each term inside the bracket is multiplied by a, keeping the original signs (+, −). |
| **2** | **C** | **−1** | A number is its own **multiplicative inverse** when a × a = 1. Only a = 1 and a = −1 satisfy this ((−1)×(−1)=1). Since the question excludes 1, the answer is −1. |
| **3** | **A** | **0.5** | If log(x+3) = log(15x−4), the **arguments must be equal**: x + 3 = 15x − 4 ⇒ 7 = 14x ⇒ x = 0.5. Both arguments (3.5) are positive, so the solution is valid. |
| **4** | **B** | **−3** | 0.00327 = 3.27 × $10^{-3}$. The **characteristic** (integer part of the logarithm) equals the power of 10, which is −3 for numbers less than 1. |
| **5** | **A** | **2x(2a + 3b + 6)** | Taking x as the **common factor**: x[(2a+3b+7)+(2a+3b+5)] = x(4a+6b+12) = 2x(2a+3b+6). |
| **6** | **B** | **a/($a^2$−1)** | Since both fractions share the **same denominator**, add the numerators directly: (2a − a)/($a^2$−1) = a/($a^2$−1). |
| **7** | **C** | **$6x^3y^8$** | √($36x^6y^{16}$) = √36 · √($x^6$) · √($y^{16}$) = 6 · $x^3$ · $y^8$. Halve each **exponent** and take the square root of the coefficient. |
| **8** | **B** | **2** | For x = 4 to satisfy 7x = k: k = 7(4) = 28. Original RHS was 30, so we must **subtract** 30 − 28 = 2. |
| **9** | **C** | **{ } (empty)** | \|x\| + 7 = 3 ⇒ \|x\| = −4. Since **absolute value can never be negative**, there is no real solution — the solution set is empty. |
| **10** | **B** | **3√31** | d = √($c^2$+$11^2$) = 20 ⇒ $c^2$ = 400 − 121 = 279 ⇒ c = √279 = √(9×31) = **3√31** (simplified radical form). |
| **11** | **C** | **(3/2, −5/2)** | Checking side lengths: (4,0)-(2,1)=√5, (2,1)-(−1,−5)=√45, (4,0)-(−1,−5)=√50. Since 5+45=50, the **hypotenuse** is between (4,0) and (−1,−5); its midpoint = (3/2, −5/2). |
| **12** | **C** | **(5, 7)** | The **centre of a circle** is the midpoint of any diameter: ((4+6)/2, (5+9)/2) = (5, 7). |

## SECTION B — Short Question Answers (11 × 3 = 33 Marks)

### Q. 2(i) — Unit 1: Laws of Exponents **3 marks**

*Simplify:* $[(-2)^3 \cdot (-2)^{-4} \cdot (-2)] \div (-2)^{-3}$

```
Numerator exponents (product rule, add): 3 + (-4) + 1 = 0        [1 mark]
So numerator = (-2)^0 = 1

Divide: 1 ÷ (-2)^-3 = (-2)^0-(-3) = (-2)^3      [1 mark]

(-2)^3 = -8                                              [1 mark]
```

**OR**

*Write as an equivalent radical/simplify: (a)* $(-27)^{1/3}$ *(b)* $(\sqrt{16})^4$ *(c)* $(\sqrt[3]{-8})^9$

```
(a) (-27)^(1/3) = ∛(-27) = -3            [1 mark]
(b) (√16)^4 = (4)^4 = 256                [1 mark]
(c) (∛-8)^9 = [(-8)^(1/3)]^9 = (-2)^9 = -512   [1 mark]
```

### Q. 2(ii) — Unit 1: Properties of Real Numbers **3 marks**

*Identify the property illustrated in each: (a)* $4 \times 1/4 = 1$ *(b)* $6+0=6$ *(c)* $2 \times (3 \times 5) = (2 \times 3) \times 5$

```
(a) 4 × 1/4 = 1  →  Multiplicative Inverse property     [1 mark]
(b) 6 + 0 = 6      →  Additive Identity property          [1 mark]
(c) 2×(3×5)=(2×3)×5  →  Associative property of multiplication  [1 mark]
```

**OR**

*State TRUE or FALSE (give a reason/counterexample if false): (a) Every real number is an irrational number. (b) Every real number is either rational or irrational.*

```
(a) FALSE. Counter-example: 5 is a real number, but 5 is
    rational, not irrational.                              [1.5 mark]
(b) TRUE. Every real number belongs to exactly one of the two
    disjoint sets: rational numbers or irrational numbers.     [1.5 mark]
```

### Q. 2(iii) — Unit 2: Laws of Logarithms **3 marks**

*(a) Expand* $\log(5pq^2/xy^3)$ *(b) Combine into a single log:* $3\log x - 5\log y$

```
(a) log(5pq^2/xy^3) = log5 + logp + 2logq − logx − 3logy [1.5 mark]
(b) 3logx − 5logy = logx^3 − logy^5 = log(x^3/y^5)     [1.5 mark]
```

**OR**

*Evaluate: (a)* $\log_9(\sqrt[3]{9})$ *(b)* $\log_{\sqrt{7}}(343)$

```
(a) log_9(9^(1/3)) = 1/3   [since log_b(b^k) = k]      [1.5 mark]
(b) √7 = 7^(1/2), 343 = 7^3
    log_(7^1/2)(7^3) = 3 ÷ (1/2) = 6                        [1.5 mark]
```

### Q. 2(iv) — Unit 2: Evaluating Logarithms **3 marks**

*If* $\log_b 2=0.3010$, $\log_b 3=0.4771$, $\log_b 5=0.6990$, *evaluate: (a)* $\log_b(6/5)$ *(b)* $\log_b(\sqrt[3]{450}/\sqrt{27})$

```
(a) log_b(6/5) = log_b6 − log_b5 = (log_b2+log_b3) − log_b5
    = (0.3010+0.4771) − 0.6990 = 0.0791                     [1.5 mark]

(b) log_b(450^(1/3)/27^(1/2)) = (1/3)log_b450 − (1/2)log_b27
    log_b450 = log_b(2×3^2×5^2) = 0.3010+2(0.4771)+2(0.6990) = 2.6532
    (1/3)(2.6532) = 0.8844
    log_b27 = 3log_b3 = 1.4313;  (1/2)(1.4313) = 0.7157(5)
    Result = 0.8844 − 0.7157 = 0.1688                          [1.5 mark]
```

**OR**

*Evaluate* $\log_b(0.024)$ *using the same given values. Show full working.*

```
0.024 = 24/1000 = (2^3×3)/(10^3) = (2^3×3)/(2^3×5^3) = 3/5^3  [1 mark]

log_b(0.024) = log_b3 − 3log_b5
             = 0.4771 − 3(0.6990)                            [1 mark]
             = 0.4771 − 2.0970 = −1.6199                    [1 mark]
```

### Q. 2(v) — Unit 4: Factorization **3 marks**

*Factorize completely: (a) $x^4 - 12x^2 + 16$ (b) $2m^4 + m^2n^2 - 3n^4$*

```
(a) x^4 − 12x^2 + 16 = (x^2−4)^2 − 4x^2      [add/subtract to form squares]
    = (x^2−4)^2 − (2x)^2 = (x^2−2x−4)(x^2+2x−4)   [1.5 mark]

(b) 2m^4 + m^2n^2 − 3n^4: treat as quadratic in m^2
    = (2m^2 + 3n^2)(m^2 − n^2) = (2m^2+3n^2)(m−n)(m+n)     [1.5 mark]
```

**OR**

*Factorize by substitution: $(y^2+2y-3)(y^2+2y+11)+48$*

```
Let u = y^2 + 2y                                                [1 mark]
(u−3)(u+11) + 48 = u^2 + 8u − 33 + 48 = u^2 + 8u + 15
= (u+3)(u+5)                                                     [1 mark]

Substitute back u = y^2+2y:
= (y^2+2y+3)(y^2+2y+5)                                           [1 mark]
```

### Q. 2(vi) — Unit 4: Sum of Squares vs Sum of Cubes **3 marks**

*$a^6+729$ as sum of squares or sum of cubes — which form factors? Factorize it.*

```
a^6+729 = (a^3)^2 + (27)^2   [sum of SQUARES — does NOT
                                     factor over real numbers]
a^6+729 = (a^2)^3 + (9)^3    [sum of CUBES — DOES factor]  [1 mark]

Use the sum-of-cubes identity is used because a sum of two
squares has no real factorization, but a sum of two cubes does.

Using x^3+y^3=(x+y)(x^2−xy+y^2), x=a^2, y=9:                    [1 mark]
= (a^2+9)(a^4 − 9a^2 + 81)                                       [1 mark]
```

**OR**

*Express $8+12t+6t^2+t^3$ as a product of three factors. Binomial or trinomial?*

```
t^3 + 6t^2 + 12t + 8   [rearranged in descending powers]

Check perfect cube: (t+2)^3 = t^3+3(t^2)(2)+3(t)(4)+8
                           = t^3+6t^2+12t+8  ✓             [2 mark]

So  8+12t+6t^2+t^3 = (t+2)(t+2)(t+2)
Each of the three factors is a BINOMIAL (t + 2).                 [1 mark]
```

### Q. 2(vii) — Unit 4: Substitution & HCF **3 marks**

*Factorize by substitution: $(2x^2-3x+6)(2x^2-3x)-55$*

```
Let u = 2x^2 − 3x                                             [1 mark]
(u+6)(u) − 55 = u^2 + 6u − 55 = (u+11)(u−5)              [1 mark]

Substitute back u = 2x^2−3x:
= (2x^2−3x+11)(2x^2−3x−5)
Factor second bracket: 2x^2−3x−5 = (2x−5)(x+1)

Final: (2x^2−3x+11)(2x−5)(x+1)                             [1 mark]
```

**OR**

*Find the HCF of $x^3+3x^2-8x-24$ and $x^3+3x^2-3x-9$ by factorization.*

```
x^3+3x^2−8x−24 = x^2(x+3) − 8(x+3) = (x+3)(x^2−8)   [1 mark]
x^3+3x^2−3x−9  = x^2(x+3) − 3(x+3) = (x+3)(x^2−3)   [1 mark]

Common factor: (x+3)   [x^2−8 and x^2−3 share no factor]

∴  HCF = (x + 3)                                                [1 mark]
```

### Q. 2(viii) — Unit 5: Fractional Linear Equation **3 marks**

*Solve for x: $(2x-11)/12 = (2x+10)/12 - [(28-2x)/4 - 1/4]$*

```
RHS, common denominator 12:
= (2x+10)/12 − 3(28−2x)/12 + 3/12
= [2x+10 − 84+6x + 3]/12 = (8x−71)/12                    [1 mark]

Equation: (2x−11)/12 = (8x−71)/12
⇒ 2x − 11 = 8x − 71                                        [1 mark]
⇒ 60 = 6x  ⇒  x = 10                                        [1 mark]
```

**OR**

*Solve: $5 - |5y+1| = -9$*

```
−|5y+1| = −14  ⇒  |5y+1| = 14                            [1 mark]

Case 1: 5y+1 = 14  ⇒  y = 13/5                               [1 mark]
Case 2: 5y+1 = −14  ⇒  y = −3                               [1 mark]

Solution set = {13/5, −3}
```

### Q. 2(ix) — Unit 5: Inequalities **3 marks**

*Solve the compound inequality: $(3/2)x \le -3$ or $(2/3)x \ge 4$, $x \in R$*

```
Inequality 1: (3/2)x ≤ −3  ⇒  x ≤ −2                     [1 mark]
Inequality 2: (2/3)x ≥ 4   ⇒  x ≥ 6                      [1 mark]

Combined (OR = union): x ≤ −2  or  x ≥ 6                    [1 mark]
```

**OR**

*Solve: $4(2y+3) - (6y-1) > 10$, $y \in R$*

```
8y + 12 − 6y + 1 > 10                                      [1 mark]
2y + 13 > 10                                                 [1 mark]
2y > −3  ⇒  y > −3/2                                         [1 mark]
```

### Q. 2(x) — Unit 7: Midpoint & Rhombus **3 marks**

*Prove diagonals of rhombus $A(-3,-1)$, $B(0,0)$, $C(1,3)$, $D(-2,2)$ bisect each other. Find diagonal lengths.*

```
Midpoint of AC = ((−3+1)/2, (−1+3)/2) = (−1, 1)
Midpoint of BD = ((0+(−2))/2, (0+2)/2) = (−1, 1)
Equal midpoints ⇒ diagonals bisect each other  ✓         [1.5 mark]

AC = √[(1−(−3))^2+(3−(−1))^2] = √(16+16) = √32 = 4√2
BD = √[(−2−0)^2+(2−0)^2] = √(4+4) = √8 = 2√2         [1.5 mark]
```

**OR**

*$A(3,4)$, $B(-1,7)$, $C(x,y)$ are collinear; B is the midpoint of AC. Find x and y.*

```
B = ((3+x)/2, (4+y)/2) = (−1, 7)                             [1 mark]

(3+x)/2 = −1  ⇒  3+x = −2  ⇒  x = −5                     [1 mark]
(4+y)/2 = 7   ⇒  4+y = 14  ⇒  y = 10                       [1 mark]
```

### Q. 2(xi) — Unit 7: Midpoint & Parallelogram **3 marks**

*Show A(0,0), B(0,5), C(4,7), D(4,2) are vertices of a parallelogram (midpoint-of-diagonals method).*

```
Diagonals are AC and BD.
Midpoint of AC = ((0+4)/2, (0+7)/2) = (2, 3.5)              [1.5 mark]
Midpoint of BD = ((0+4)/2, (5+2)/2) = (2, 3.5)              [1 mark]

Both diagonals share the same midpoint ⇒ they bisect each
other ⇒ ABCD is a parallelogram.                             [0.5 mark]
```

**OR**

*P(2,3) is the midpoint of A(x,4) and B(−3,y). Find x and y.*

```
((x+(−3))/2, (4+y)/2) = (2, 3)                              [1 mark]

(x−3)/2 = 2  ⇒  x−3 = 4  ⇒  x = 7                         [1 mark]
(4+y)/2 = 3   ⇒  4+y = 6  ⇒  y = 2                          [1 mark]
```

## SECTION C — Long Question Answers (4 × 5 = 20 Marks)

### Q. 3(i) — Unit 2: Advanced Logarithm Evaluation **5 marks**

*Given $\log_b 2=0.3010$, $\log_b 3=0.4771$, $\log_b 5=0.6990$, evaluate $\log_b 0.0036$.*

```
0.0036 = 36/10000 = (2^2×3^2)/(2^4×5^4) = 3^2/(2^2×5^4)     [1 mark]

log_b(0.0036) = 2log_b3 − 2log_b2 − 4log_b5                  [1 mark]
              = 2(0.4771) − 2(0.3010) − 4(0.6990)
              = 0.9542 − 0.6020 − 2.7960                       [1 mark]
              = −2.4438

This is negative with both characteristic and mantissa
negative — mantissa must never be negative, so add and
subtract 3 (next integer above the magnitude 2.4438):          [1 mark]
  log_b(0.0036) = −2.4438 + 3 − 3 = 0.5562 − 3

∴  log_b(0.0036) = bar-3 . 5562  (characteristic −3,
    mantissa 0.5562)                                            [1 mark]
```

**OR**

*Evaluate using logarithm tables: 7th root of 129.4 ÷ cube root of 27.37*

```
Let x = 129.4^(1/7) ÷ 27.37^(1/3)
log x = (1/7)log(129.4) − (1/3)log(27.37)                    [1 mark]

log(129.4) = 2 + log(1.294) ≈ 2.1119   (from log table)      [1 mark]
log(27.37) = 1 + log(2.737) ≈ 1.4373   (from log table)      [1 mark]

log x = (1/7)(2.1119) − (1/3)(1.4373)
      = 0.3017 − 0.4791 = −0.1774
      = bar-1 . 8226  (adjust for positive mantissa)             [1 mark]

antilog(0.8226) ≈ 6.647  ⇒  x ≈ 6.647 × 10^−1 ≈ 0.6647   [1 mark]
```

### Q. 3(ii) — Unit 4: Grouping + Sum/Difference of Cubes & Squares 5 marks

*Factorize completely:* $x^3p^2 - 8y^3p^2 - 4x^3q^2 + 32y^3q^2$

```
Group in pairs:
= p^2(x^3 − 8y^3) − 4q^2(x^3 − 8y^3)                      [1 mark]
= (x^3 − 8y^3)(p^2 − 4q^2)          [common factor]        [1 mark]

Factor x^3−8y^3 = (x)^3−(2y)^3 = (x−2y)(x^2+2xy+4y^2)       [2 mark]
Factor p^2−4q^2 = (p−2q)(p+2q)         [difference of squares] [1 mark]

∴  Result = (x−2y)(x^2+2xy+4y^2)(p−2q)(p+2q)
```

**OR**

*Simplify (all denominators non-zero):* $[(x^2+x-2)/(x^2-x-20)] \times [(x^2+5x+4)/(x^2-x)] \div \{[(x^2+3x+2)/(x^2-2x-15)] \times [(x+3)/x^2]\}$

```
Factorize everything:
x^2+x−2=(x+2)(x−1)   x^2−x−20=(x−5)(x+4)
x^2+5x+4=(x+4)(x+1)  x^2−x=x(x−1)
x^2+3x+2=(x+2)(x+1)  x^2−2x−15=(x−5)(x+3)              [1.5 mark]

First product (numerator group), cancel (x−1),(x+4):
= (x+2)(x+1) / [(x−5)x]                                    [1 mark]

Divisor group, cancel (x+3):
= (x+2)(x+1) / [(x−5)x^2]                                  [1 mark]

Divide (multiply by reciprocal):
= [(x+2)(x+1)/((x−5)x)] × [(x−5)x^2/((x+2)(x+1))]
= x^2/x = x                                                 [1.5 mark]

∴  Result = x
```

### Q. 3(iii) — Unit 5: Radical Equation & Compound Inequality 5 marks

*(a) Solve:* $\sqrt{(a-1/2)} = \sqrt{(2a/5+2/5)}$ *[3] (b) Three times a number y minus 18 is less than 12 or greater than 39. Find y. [2]*

```
(a) Square both sides: a − 1/2 = 2a/5 + 2/5                 [1 mark]
    Multiply by 10: 10a − 5 = 4a + 4
    6a = 9  ⇒  a = 3/2                                       [1 mark]
    Check: LHS=√(3/2−1/2)=√1=1; RHS=√(3/5+2/5)=√1=1 ✓  [1 mark]

(b) 3y − 18 < 12  or  3y − 18 > 39
    3y < 30  ⇒  y < 10        3y > 57  ⇒  y > 19        [1 mark]
    ∴  y < 10  or  y > 19                                  [1 mark]
```

**OR**

*(a) Solve:* $[\sqrt{(5x-4)}-4]/10 = -1$ *(state if extraneous) [3] (b) Solve:* $1.3x-0.2 = 0.3x-1.5$ *[2]*

```
(a) √(5x−4) − 4 = −10  ⇒  √(5x−4) = −6            [1.5 mark]
    A principal square root can never be negative, so
    this equation has NO real solution — solution
    set = ∅ (extraneous / no solution).                    [1.5 mark]

(b) 1.3x − 0.3x = −1.5 + 0.2                               [1 mark]
    1.0x = −1.3  ⇒  x = −1.3                             [1 mark]
```

**Q. 3(iv) — Unit 7: Rectangle & Right Triangles / Midsegment Theorem** **5 marks**

*Show A(2,−1), B(8,−1), C(8,3), D(2,3) are vertices of a rectangle. Show triangles ABC and ABD are right triangles.*

```
AB=√[(8−2)^2+0^2]=6   BC=√[0^2+(3−(−1))^2]=4
CD=√[(2−8)^2+0^2]=6   DA=√[0^2+(−1−3)^2]=4
Opposite sides equal, all sides axis-aligned (right angles)
⇒ ABCD is a rectangle.                                       [2 mark]

Diagonal AC=√[(8−2)^2+(3−(−1))^2]=√(36+16)=√52
ΔABC: AB^2+BC^2 = 36+16 = 52 = AC^2 ⇒ right ∠ at B  ✓    [1.5 mark]

Diagonal BD=√[(2−8)^2+(3−(−1))^2]=√(36+16)=√52
ΔABD: AB^2+AD^2 = 36+16 = 52 = BD^2 ⇒ right ∠ at A  ✓    [1.5 mark]
```

**OR**

*ΔABC: A(3,7), B(−4,0), C(4,0). D, E are midpoints of AB, AC. Find D,E; d(D,E) and d(B,C); compare.*

```
D = midpoint AB = ((3−4)/2,(7+0)/2) = (−0.5, 3.5)
E = midpoint AC = ((3+4)/2,(7+0)/2) = (3.5, 3.5)                [1.5 mark]

d(D,E) = √[(3.5−(−0.5))^2+0^2] = √16 = 4                [1 mark]
d(B,C) = √[(4−(−4))^2+0^2] = √64 = 8                    [1 mark]

d(B,C) = 2 × d(D,E)  ⇒  DE is parallel to BC and equal to
half of BC (Midsegment / Midpoint Theorem confirmed).           [1.5 mark]
```



**— End of Answer Key —**